

Individual Assignment: News Sentiment Trading Strategy Using Natural Language Processing**Objective**

*1-minute interval - Every Minutes (Refers to keeping dataset)

To formulate an AI/ML-based news-sentiment trading strategy on a 1-minute interval Bursa Malaysia stock. Students must demonstrate the ability to translate a sentiment signal into a defensible, risk-managed trading rule, evaluate its performance across multiple holding horizons, and interpret the results with genuine financial reasoning.

This IA must use FINBERT

Mandate

A quantitative research desk has requested an evaluation of a sentiment-driven trading rule on a single Bursa Malaysia counter. You are required to:

- i. Select **ONE (1)** Bursa Malaysia listed stock with sufficiently complete 1 minute interval closing price data for the exact window **13 April 2026 to 8 July 2026**.
- ii. Confirm the selected stock recorded **positive average daily return** over this period (the mean of day-to-day percentage price changes across the window must be greater than zero).
- iii. News headline data for the same period will be provided; you are required to formulate the sentiment-to-trade decision rule applied to it.
- iv. Execute, evaluate, and defend the resulting strategy's performance, including its risk controls.

Required:

*Stock must be different from each classmate

1. Identify **ONE (1)** Bursa Malaysia stock with sufficiently complete 1 minute closing price data for 13 April 2026 to 8 July 2026 that records a positive average daily return over the window.
2. Prepare your selected stock's raw price dataset and the provided news headline dataset for the same window, with missing or illiquid minute observations treated appropriately.
3. Explain the key descriptive statistics of your selected stock, including average daily return, skewness, kurtosis, price range, number of trading days, missing observations and liquidity issues.
4. Formulate a defensible Minimum Daily Score conviction threshold and Holding Days list based on your stock's data behaviour and the nature of the news signal.
5. Alter the Back-testing Assumptions and Parameters section to realistic figures for your stock's price level, liquidity, Malaysian retail brokerage fee and the 3rd quarter 2026 OPR based risk free rate.
6. Alter the supplied FINBERT configuration by explaining the roles of model name and batch size, comparing at least one alternative batch size, and defending the final setting using dataset size, processing speed, memory usage and error free execution.
7. Compare performance across every holding horizon tested using win rate, average return per trade, cumulative return, ending equity, maximum drawdown, Sharpe ratio and profit factor, then synthesise the results to identify the best horizon and configuration.
8. Explain at least **ONE (1)** genuine anomaly in your own results, then relate the findings to financial theory, including market efficiency, information spillover and transaction cost drag, and generalise the implications for news based intraday trading.

Report Structure and Formatting

- i. Professional, consistently formatted academic report. All tables, figures and equations labelled and referenced in-text.
- ii. APA 7th edition formatting for all in-text citations and references.
- iii. The report must include: (i) clean, stage-labelled Python code covering data preparation, sentiment scoring, strategy execution, and evaluation; (ii) price/sentiment plots and a multi-horizon performance table; (iii) a structured analytical discussion linking results back to the defended parameters.

Submission Files

- i. Final individual report: approximately 2,000 words, maximum 10 pages excluding cover page, references and appendices. Format: PDF.
- ii. Complete Jupyter Notebook (.ipynb): must run end-to-end without error and reproduce every reported figure exactly.
- iii. Raw and cleaned datasets (price) in .csv or .xlsx format, covering the exact required window.
- iv. Item ii. and iii. must be submitted in **ONE (1)** zipped folder.

Referencing

At least **FIVE (5)** credible academic or professional sources, APA 7th edition, supporting the sentiment methodology, risk-management approach, and financial interpretation.

Cover Page

Name, Student ID, module code, and assignment title. Format: [Name, TP No., RBA IA 062026].

Important Notes

- i. All strategy formulation, execution, and interpretation must be based on your own selected stock, your own dataset, and your own empirical results. Generic explanations not traceable to your submitted code and data will be treated as unsupported.
- ii. Every reported number must be reproducible from your submitted notebook. Mismatches between the report and notebook output will be treated as a serious inconsistency and may be investigated as possible fabricated or unsupported results.
- iii. Parameter choices must be defended using statistics computed from your own data. Round, generic, or textbook-default values without justification will be penalised under Criterion 3 or Criterion 4, depending on the parameter involved.
- iv. Maintain a professional, analytical tone, as if reporting to a quantitative research desk.
- v. A similarity index above 20% (Turnitin or equivalent) results in a minimum 10% deduction, subject to further academic review. Submissions with high AI detection indicators, unsupported generic interpretation, or inability to defend reported figures may be flagged for academic review.

[Grand Total: 100 marks]

RUBRIC

Criterion	Weight (%)	Band 5 Excellent	Band 4 Strong	Band 3 Satisfactory	Band 2 Weak	Band 1 Poor
1. Stock Selection & Data Authenticity	10	Correct Bursa Malaysia stock, exact 1-min data 13/4/2026–8/7/2026, positive average daily return correctly computed and verified with timestamped raw export.	Correct stock and date range with positive average daily return confirmed. Minor gaps in verification.	Correct date range but average daily return calculation or verification trail incomplete.	Date range partially incorrect, or growth condition checked using endpoint prices rather than average daily return.	Wrong stock, wrong date range, or growth condition not met/fabricated.
2. Descriptive Statistics & Data Characterisation	15	All required statistics, including average daily return, skewness, kurtosis, price range, trading days, missing observations and liquidity issues, are correctly computed.	All statistics present and mostly correctly explained; minor interpretive gaps.	Statistics computed but explanation largely descriptive, not interpretive.	Statistics incomplete or partially incorrect.	Statistics missing, incorrect, or not linked to the selected stock's actual data.
3. Strategy Formulation: Threshold & Horizon Design	20	Minimum Daily Score and Holding Days list clearly formulated and justified against the stock's own data behaviour and the nature of the news signal.	Thresholds formulated correctly with minor justification gaps.	Thresholds present but justification largely generic or textbook derived.	Thresholds vague, inconsistently applied, or weakly justified.	No coherent threshold/horizon formulation; CLO3 not evidenced.
4. Back-Testing Assumptions & FinBERT Configuration Defence	20	Capital, cost and risk-free assumptions altered to realistic, sourced figures; FinBERT model and batch size roles explained, at least one alternative batch size compared, and final setting defended on dataset size, speed, memory and error-free execution.	Assumptions realistic and FinBERT settings mostly defended; minor gaps in comparison or sourcing.	Assumptions or FinBERT settings altered but defence largely generic, with limited or no real comparison.	Assumptions or FinBERT settings left mostly at default with weak or no defence.	Assumptions unrealistic/un sourced; FinBERT configuration undefended or code fails to execute.

RUBRIC (cont.)

Criterion	Weight (%)	Band 5 Excellent	Band 4 Strong	Band 3 Satisfactory	Band 2 Weak	Band 1 Poor
5. Performance Evaluation & Multi-Horizon Comparison	15	Full metric set computed correctly across every horizon tested; results compared and synthesised to identify the best-performing horizon and configuration with clear support.	Metrics mostly correct and compared; synthesis of best configuration present with minor gaps.	Metrics computed but cross-horizon comparison or synthesis superficial.	Metrics incomplete, miscalculated, or comparison largely absent.	Metrics absent, incorrect, or not linked to the strategy tested.
6. Critical Reflection & Financial Interpretation	15	At least one genuine anomaly correctly explained; findings clearly related to market efficiency, information spillover and transaction cost drag, with implications properly generalised.	Anomaly explained and findings related to theory with minor gaps.	Anomaly identified but explanation shallow; theory linkage generic.	Anomaly noted without meaningful explanation; weak theory linkage.	No anomaly identified; reflection generic or absent; no theory linkage.
7. Report Structure, Referencing & Academic Writing	5	Professional structure, correctly labelled tables/figures, APA 7th referencing, analytical academic tone throughout.	Minor formatting or referencing issues.	Basic structure; inconsistent referencing.	Disorganised layout; referencing largely missing.	Incomplete structure or evidence of plagiarism.

Bonus Marks (Up to +5)

Awarded only for clear originality and effort beyond the required scope:

- +2 for a genuinely novel refinement to the strategy (e.g. confidence-weighted position sizing, an asymmetric TP/SL tested and compared).
- +2 for integrating a real, dated macroeconomic or company-specific event into the interpretation of a specific trade or anomaly.
- +1 for exceptionally polished formatting and report flow.

Penalty Deductions (Up to -10 marks)

- -2 to -10 for AI-like generic phrasing or interpretation not grounded in the student's own reported figures.
- -3 to -5 if reported results do not match the submitted notebook output.
- -5 to -10 if template code or parameters are reused without demonstrated understanding or adaptation to the student's own data.

END OF QUESTION PAPER